DATA ENGINEERING ASSIGNMENT

Apache Pig: Online Shopping Product Review Analysis

Filtering invalid ratings and calculating average rating by product category

Tool: Apache Pig (Pig Latin)

Storage: HDFS (Hadoop Distributed File System)

Dataset: Online shopping product reviews (CSV)

Execution mode shown in screenshots: Hadoop MapReduce

1. Aim

To use Apache Pig to process an online-shopping product-review dataset stored in HDFS, remove the CSV header and invalid ratings, group valid reviews by product category, and calculate the average rating for each category.

2. Problem Statement

An e-commerce company receives product reviews across categories such as Electronics, Home Appliances, Fashion, Books, and Beauty. Some review ratings may be missing or invalid. Create a Pig Latin script that loads the review data from HDFS, removes null ratings and ratings below 1.0, groups records by product category, and computes the average rating per category.

3. Software and Tools Used

- Apache Hadoop and HDFS — store and access the CSV input and output files.
- Apache Pig — express data processing using Pig Latin.
- Pig Latin — script language used for LOAD, FILTER, FOREACH, GROUP, AVG, and STORE operations.
- Linux terminal and nano editor — create the CSV file and Pig script and run commands.

4. Dataset: Online Shopping Product Reviews

The dataset contains 15 product reviews. The rating column is stored as text initially so that the script can explicitly exclude the literal value 'null' before converting valid ratings to numbers.

review_id	product_category	product_name	rating
R501	Electronics	Wireless_Earbuds	4.8
R502	Electronics	Bluetooth_Speaker	4.3
R503	Electronics	Smartwatch	0.5
R504	Home_Appliances	Air_Fryer	4.6
R505	Home_Appliances	Robot_Vacuum	4.2
R506	Home_Appliances	Electric_Kettle	null
R507	Fashion	Running_Shoes	4.5
R508	Fashion	Denim_Jacket	4.0
R509	Fashion	Backpack	0.0
R510	Books	Data_Science_Book	4.7
R511	Books	Programming_Guide	4.1
R512	Books	Machine_Learning_Book	4.4
R513	Beauty	Face_Cream	3.9
R514	Beauty	Shampoo	4.3
R515	Beauty	Hair_Serum	4.6

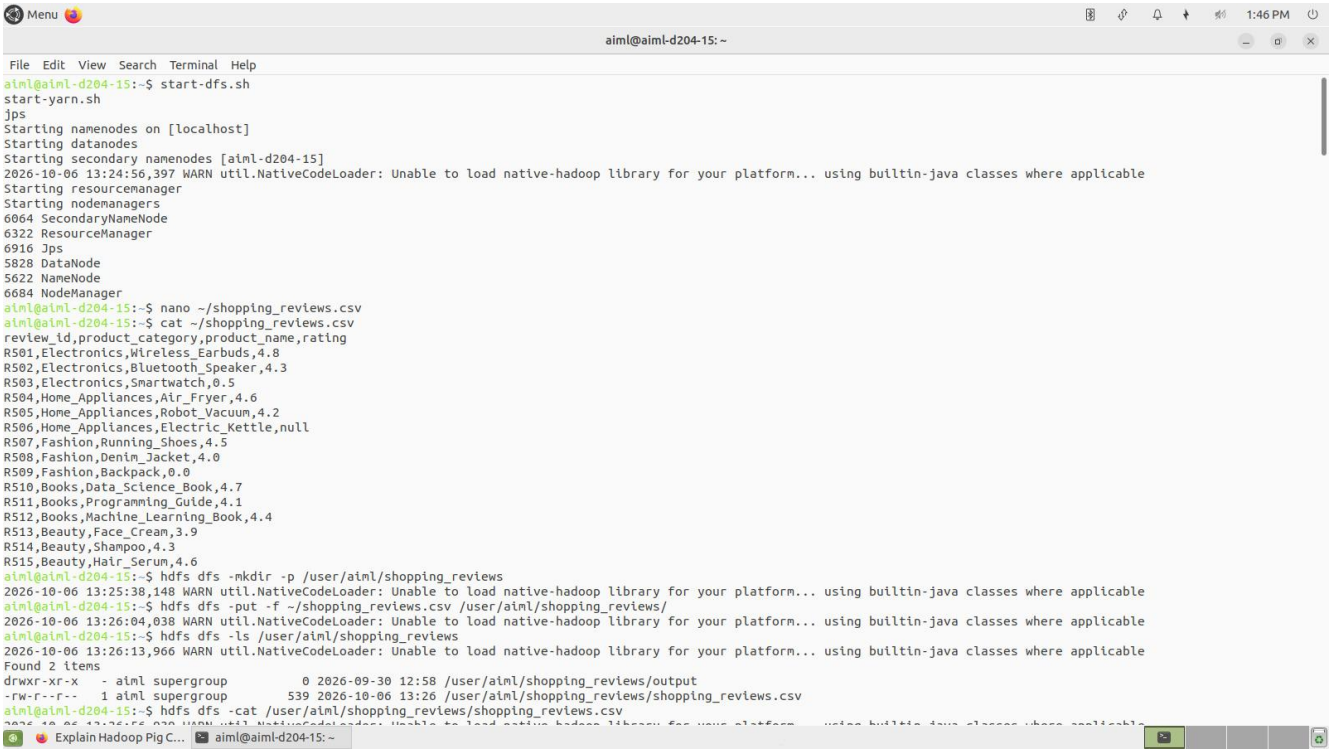
Invalid values included for testing: R503 has rating 0.5, R509 has rating 0.0, and R506 has rating null. All three should be excluded from the average calculations.

5. Step-by-Step Execution Commands

Step 1 — Start Hadoop services

```
start-dfs.sh
start-yarn.sh
jps
```

The start scripts start HDFS and YARN services. The jps command lists Java processes, which can help confirm that Hadoop daemons such as NameNode, DataNode, ResourceManager, and NodeManager are running. The exact processes shown depend on the installation.



Step 2 — Create and inspect the CSV dataset

```
nano ~/shopping_reviews.csv
cat ~/shopping_reviews.csv
```

nano opens a text editor to create or edit the CSV file. The cat command prints the file contents so the dataset can be checked before uploading it to HDFS.

Step 3 — Create an HDFS directory and upload the CSV

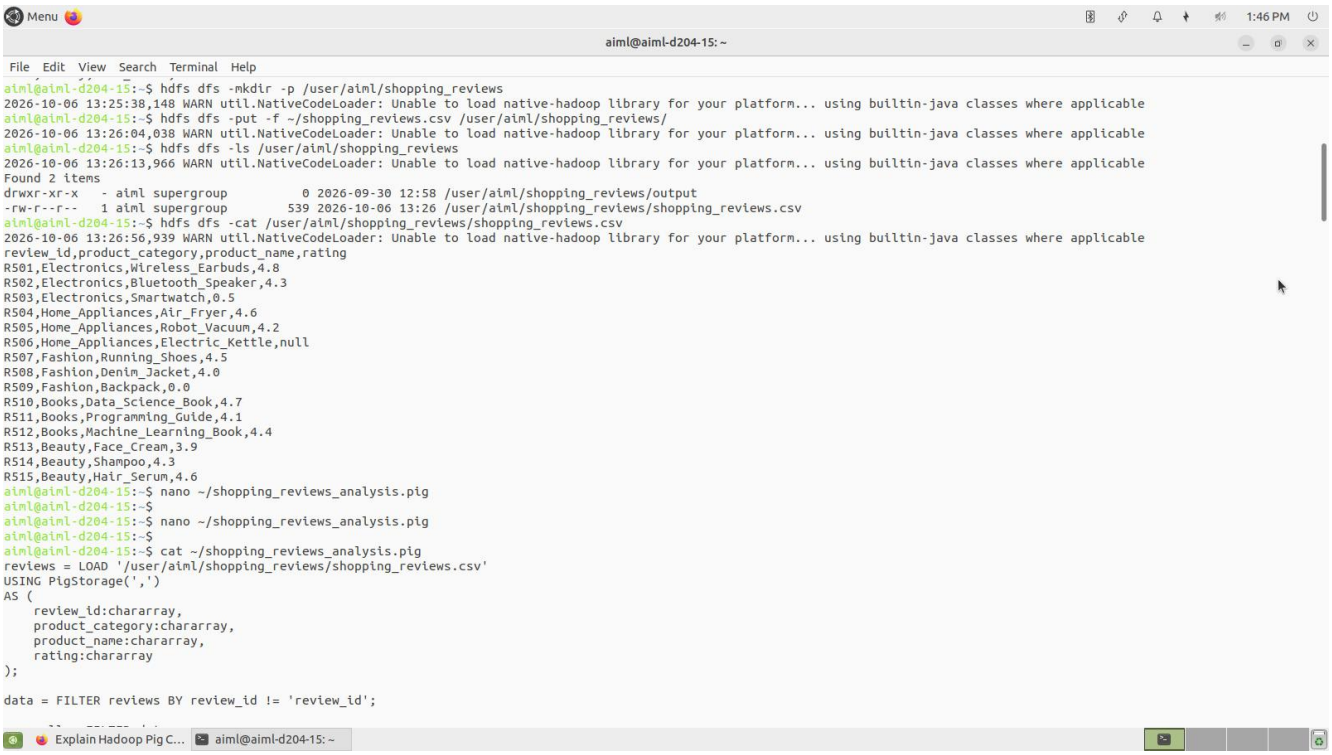
```
hdfs dfs -mkdir -p /user/aiml/shopping_reviews
hdfs dfs -put -f ~/shopping_reviews.csv /user/aiml/shopping_reviews/
```

mkdir -p creates the HDFS directory if it does not already exist. The -put command copies the local CSV file into HDFS; -f allows an existing destination file to be overwritten.

Step 4 — Verify the uploaded file

```
hdfs dfs -ls /user/aiml/shopping_reviews
hdfs dfs -cat /user/aiml/shopping_reviews/shopping_reviews.csv
```

The -ls command lists files and directories in HDFS. The -cat command displays the uploaded CSV contents, confirming that the file is available at the expected HDFS path.



6. Apache Pig Script

Create the script file with:

```
nano ~/shopping_reviews_analysis.pig
```

Use the following Pig Latin code:

```
reviews = LOAD '/user/aiml/shopping_reviews/shopping_reviews.csv'
USING PigStorage(',')
AS (
    review_id:chararray,
    product_category:chararray,
    product_name:chararray,
    rating:chararray
);

data = FILTER reviews BY review_id != 'review_id';

non_null = FILTER data
BY rating IS NOT NULL AND rating != 'null';

typed_data = FOREACH non_null GENERATE
    review_id,
    product_category,
    product_name,
    (float)rating AS rating;

valid_reviews = FILTER typed_data BY rating >= 1.0;
```

```
grouped_reviews = GROUP valid_reviews BY product_category;

average_ratings = FOREACH grouped_reviews GENERATE
    group AS product_category,
    AVG(valid_reviews.rating) AS average_rating;

STORE average_ratings
INTO '/user/aiml/shopping_reviews/output'
USING PigStorage(',');
```

7. Explanation of the Pig Script

- **LOAD —** Reads the CSV file from the HDFS path.
- **PigStorage(',') —** Tells Pig that commas separate the fields in the CSV.
- **AS (...) —** Defines the four columns and their initial data types. rating is chararray because the CSV contains the text 'null'.
- **FILTER review_id != 'review_id' —** Removes the header row so the column names are not treated as a review.
- **FILTER rating IS NOT NULL AND rating != 'null' —** Removes missing ratings, including the literal text null in the CSV.
- **FOREACH ... GENERATE —** Creates a new version of each remaining row and casts rating to float so it can be used in arithmetic.
- **FILTER rating >= 1.0 —** Keeps ratings of 1.0 or above and removes 0.5 and 0.0.
- **GROUP ... BY product_category —** Places reviews with the same product category into groups.
- **AVG(valid_reviews.rating) —** Calculates the mean rating inside each category group.
- **STORE ... INTO ... —** Writes the category averages to the HDFS output directory.

8. Run the Script and Check the Output

First remove the old output directory if it exists, because Pig generally cannot write to an existing output directory:

```
hdfs dfs -rm -r -f /user/aiml/shopping_reviews/output
```

Run the Pig script in MapReduce mode:

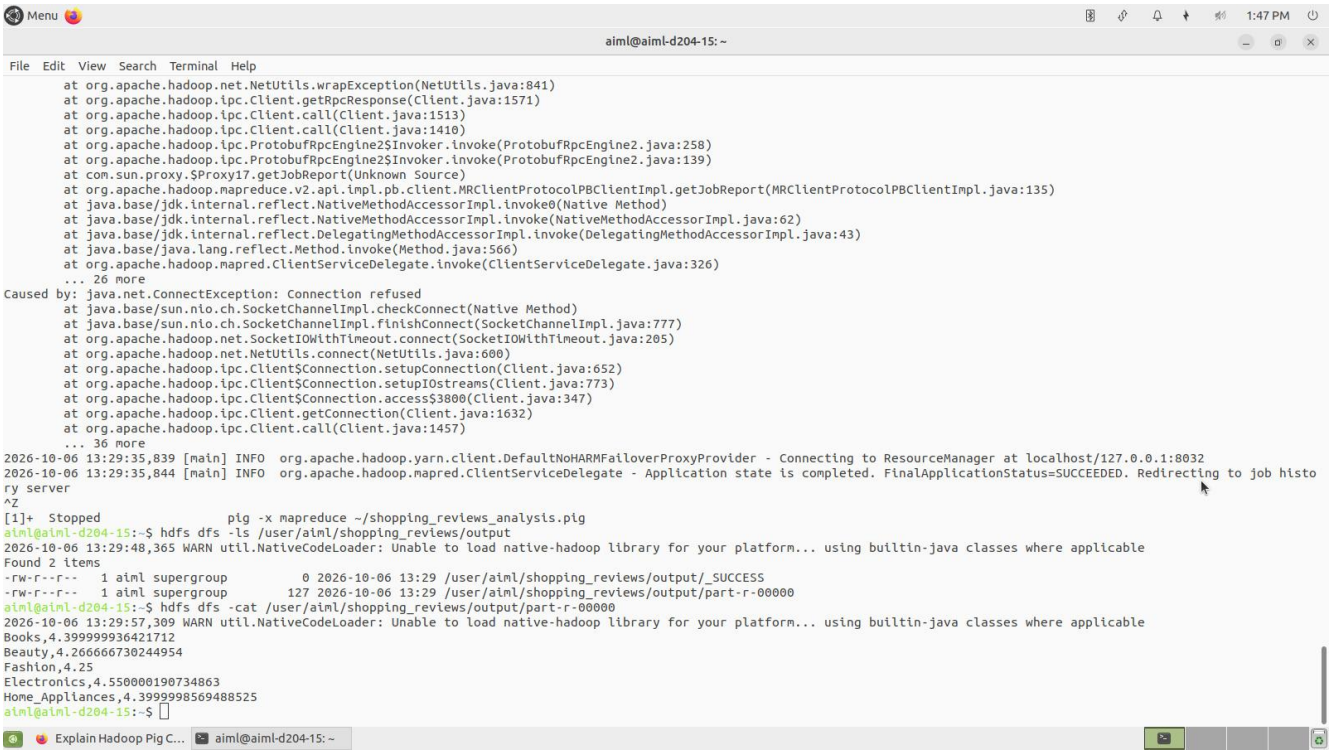
```
pig -x mapreduce ~/shopping_reviews_analysis.pig
```

Pig parses the script, builds an execution plan, and submits the processing job to Hadoop. Log lines and warnings may appear in the terminal. The important check is whether the job completes and the output directory is created.

List the output directory and display the result:

```
hdfs dfs -ls /user/aiml/shopping_reviews/output
hdfs dfs -cat /user/aiml/shopping_reviews/output/part-r-00000
```

The output directory normally contains a part file and a _SUCCESS marker. The part file contains the category averages. The output order can vary because distributed processing does not guarantee a particular category order.

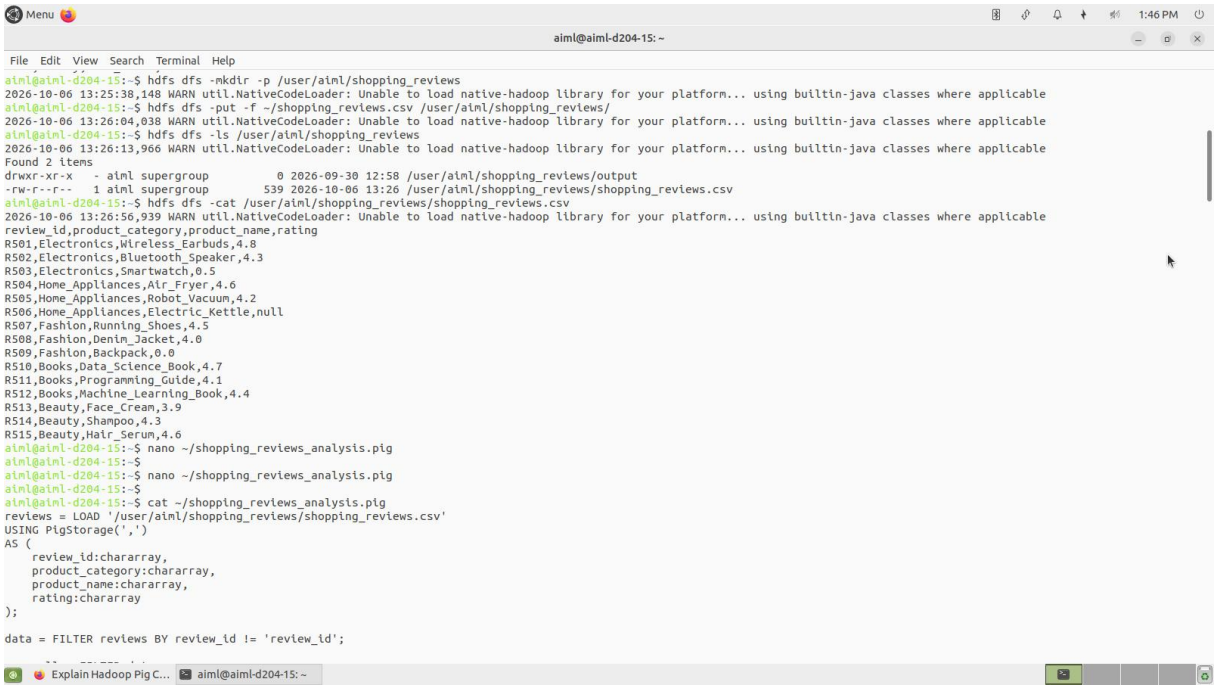


9. Expected Output and Calculation

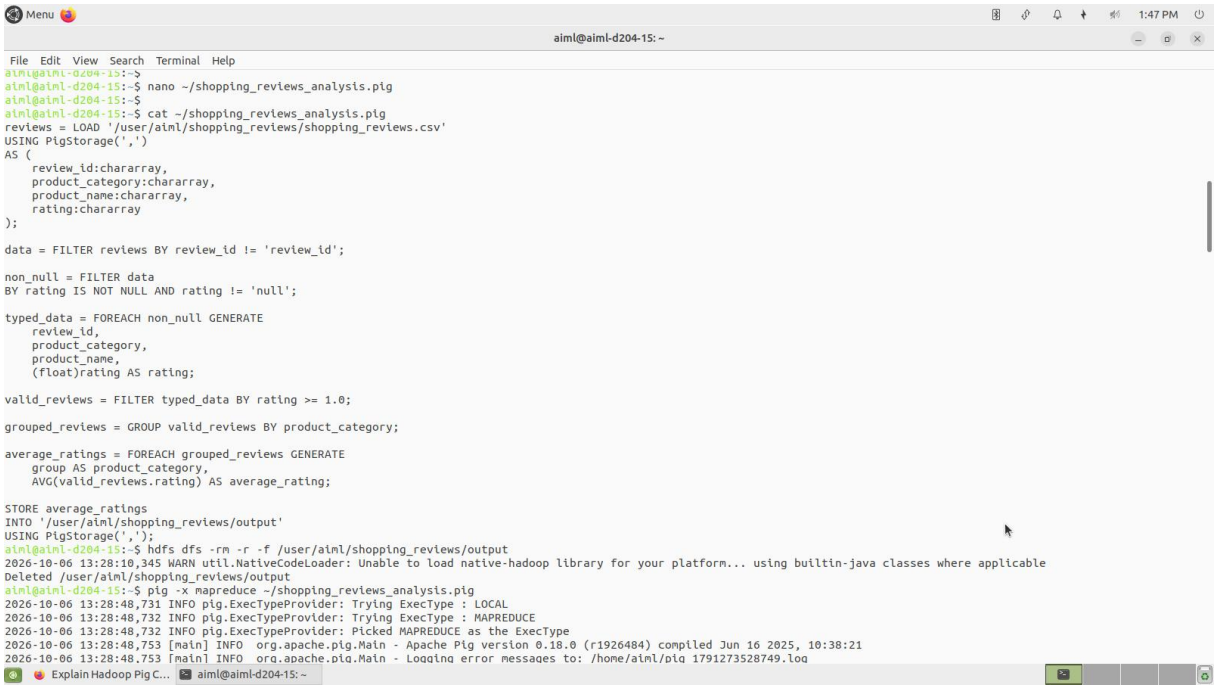
The script starts with 15 review records. It removes one null rating and two ratings below 1.0, leaving 12 valid reviews. It then calculates one average for each of the five categories.

Product category	Valid ratings	Average rating (approx.)
Books	4.7, 4.1, 4.4	4.40
Beauty	3.9, 4.3, 4.6	4.2667
Electronics	4.8, 4.3	4.55
Fashion	4.5, 4.0	4.25
Home_Appliances	4.6, 4.2	4.40

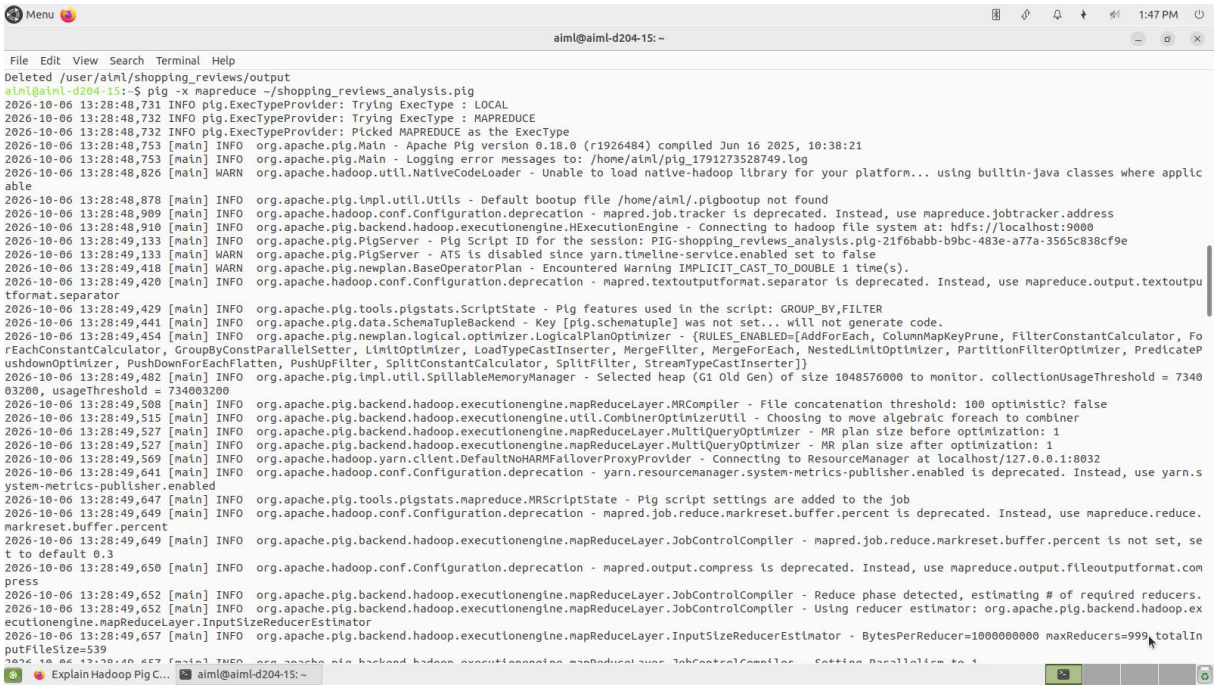
The terminal may display more decimal places than the rounded values shown above. For example, the average for Beauty may appear as approximately 4.2666667 and Electronics as approximately 4.5500002 due to floating-point representation.



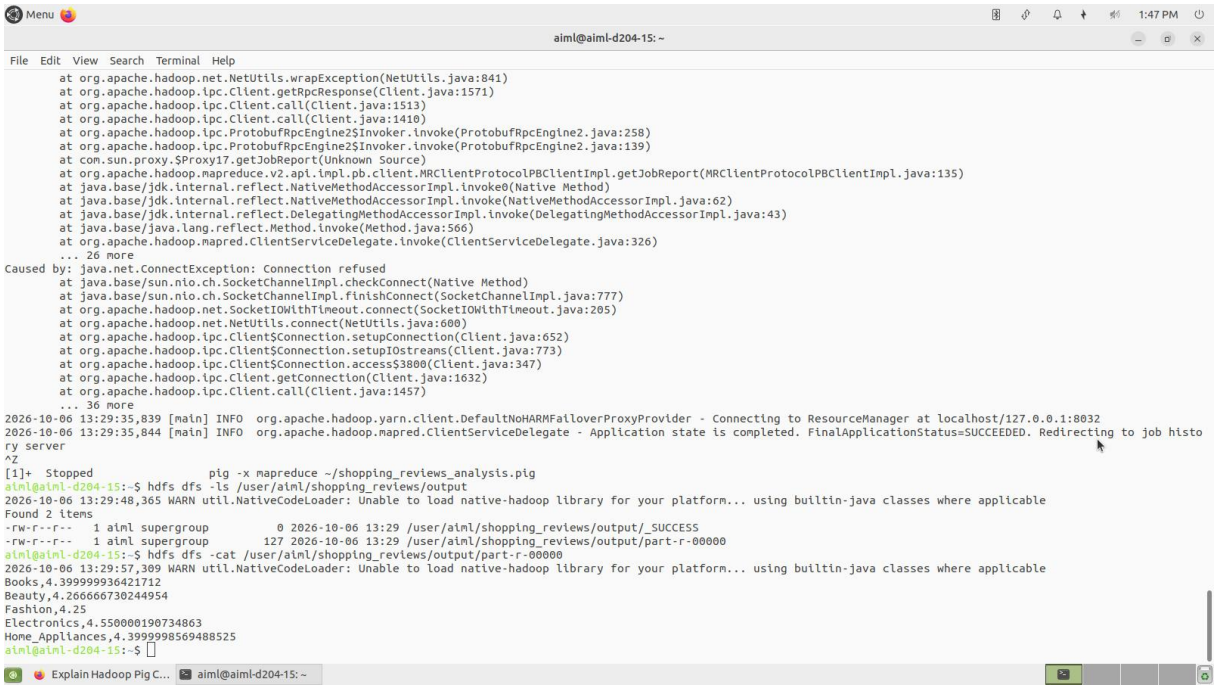
Screenshot 1 — HDFS setup, CSV upload, dataset verification, and Pig script contents (first part).



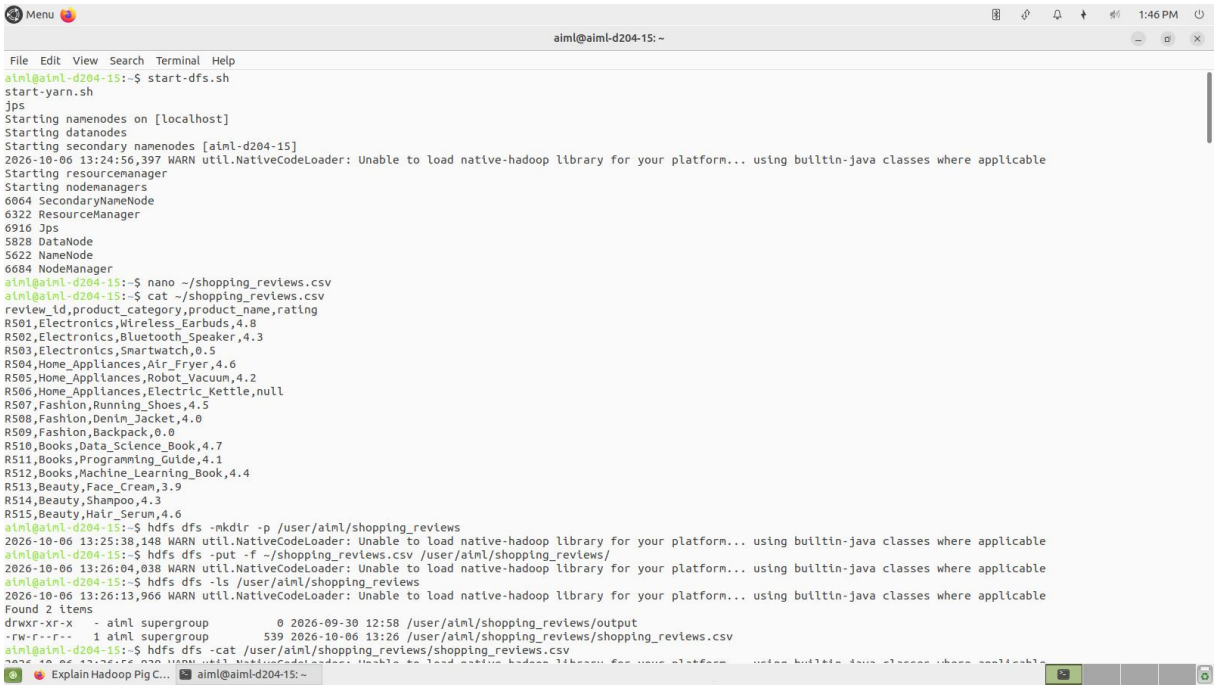
Screenshot 2 — Complete Pig Latin script and start of the execution command.



Screenshot 3 — Pig execution logs showing the script being submitted to the Hadoop/MapReduce environment.



Screenshot 4 — Execution completion/output verification area, including the HDFS output listing and category averages.



Screenshot 5 — Hadoop services started and the online-shopping review CSV dataset displayed in the terminal.